

BHISHO, South Africa

WMO: 687520

Lat: 32.894S

Lon: 27.287E

Elev: 1936

StdP: 13.70

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.3	43.0	18.5	15.0	64.7	22.5	18.2	64.3	33.4	53.2	29.2	55.5	10.4	260	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.8	91.7	67.3	88.1	67.2	84.8	66.4	73.2	84.1	71.6	81.8	70.1	79.2	10.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	119.5	77.1	68.6	112.8	75.1	67.3	107.9	73.8	38.2	84.3	36.6	82.2	35.3	79.4	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.0	23.2	20.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	34.5	77.5	2.1	2.4	33.0	79.2	31.8	80.6	30.6	82.0	29.1	83.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	69.5	70.2	68.0	64.6	61.7	57.7	57.2	58.7	60.3	61.6	63.7	67.5
	DBStd	7.46	5.53	5.46	5.80	6.28	5.59	5.73	6.37	6.83	6.64	6.59	5.74	5.71
	HDD50	23	0	0	0	0	0	5	10	5	3	0	0	0
	HDD65	1436	19	12	33	83	136	226	254	222	173	147	91	40
	CDD50	4895	604	566	558	438	364	237	232	274	310	360	411	541
	CDD65	834	157	158	127	71	35	8	11	26	31	42	52	116
	CDH74	6235	1073	1008	842	487	250	78	97	300	355	423	428	894
	CDH80	2027	359	348	295	147	46	5	7	90	124	156	133	317
(14) Wind	WSAvg	10.3	10.8	10.2	9.6	9.3	9.1	10.1	10.0	10.7	10.8	11.2	11.3	10.8
(15) Precipitation	PrecAvg	28.7	3.6	3.1	3.4	2.0	1.0	1.0	1.0	1.5	1.9	2.9	3.6	3.7
	PrecMax	35.7	6.3	5.7	9.3	5.0	4.5	6.3	4.5	6.1	8.1	8.0	9.3	8.9
	PrecMin	20.6	1.4	1.4	0.6	0.5	0.2	0.1	0.0	0.2	0.3	0.6	0.4	0.8
	PrecStd	3.7	1.5	1.0	1.7	1.2	1.1	1.3	1.2	1.3	1.8	1.9	2.0	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.4	94.3	94.5	90.3	84.5	80.2	80.5	88.3	90.8	93.1	91.0	93.9
		MCWB	70.8	71.4	68.4	63.8	61.6	55.5	54.0	58.6	60.2	63.0	63.8	69.4
	2%	DB	88.6	88.2	87.5	84.9	80.6	76.3	76.8	81.3	82.8	85.3	83.6	87.8
		MCWB	71.2	71.0	67.1	63.0	59.9	53.7	53.0	55.4	60.0	62.9	66.1	69.7
	5%	DB	83.9	84.0	82.3	79.8	76.3	72.8	72.9	76.5	77.2	78.6	78.8	83.2
		MCWB	70.0	70.6	67.7	63.2	58.0	52.7	51.9	54.3	58.7	62.0	65.2	68.1
	10%	DB	80.0	80.0	78.1	75.5	71.8	68.5	68.4	71.6	72.0	73.7	74.7	78.7
		MCWB	68.2	69.0	66.4	62.5	56.8	51.1	50.8	52.9	57.4	60.5	63.6	66.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.2	75.8	72.5	69.3	66.7	59.0	58.0	63.0	67.2	68.9	71.5	74.5
		MCDB	86.1	86.4	84.5	77.0	75.1	67.0	65.9	75.3	79.5	81.4	81.7	87.7
	2%	WB	72.8	73.4	70.4	67.3	62.2	56.6	56.3	60.1	63.1	65.8	68.3	71.5
		MCDB	84.0	83.1	80.7	76.8	71.2	66.2	65.7	70.6	74.1	76.6	78.5	82.5
	5%	WB	71.0	71.6	68.8	65.6	60.4	55.2	54.9	58.3	61.0	63.9	66.4	69.7
		MCDB	80.6	80.8	78.2	74.3	69.3	66.1	65.1	67.9	70.8	74.3	75.2	79.2
	10%	WB	69.3	70.0	67.3	63.9	58.9	53.7	53.4	56.4	59.1	62.1	64.6	68.0
		MCDB	77.6	77.9	75.8	71.5	68.4	64.1	63.3	64.5	68.4	71.2	71.9	76.0
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	16.8	17.3	17.7	17.9	18.9	18.5	19.7	20.2	18.2	17.6	17.2
		MCDBR	25.4	24.5	25.6	24.5	24.3	24.4	24.4	28.4	30.0	27.9	26.1	27.2
	5% WB	MCWBR	10.5	9.9	9.9	9.1	10.0	10.1	9.2	9.9	11.7	11.5	11.3	11.6
		MCWBR	22.3	22.4	21.7	20.6	19.6	20.2	19.1	21.0	24.4	21.8	21.8	22.4
(40) Clear-Sky Solar Irradiance	taub		0.380	0.380	0.364	0.335	0.317	0.302	0.306	0.356	0.384	0.393	0.372	0.376
		taud	2.433	2.441	2.460	2.510	2.531	2.542	2.524	2.376	2.295	2.315	2.412	2.437
	Ebn at Noon		304	298	291	282	272	268	273	270	279	290	304	306
		Edh at Noon	39	37	34	29	25	23	25	32	39	42	39	39
(44) All-Sky Solar Radiation	RadAvg		2071	1901	1612	1283	1019	911	977	1205	1532	1774	2024	2129
	RadStd		117	103	125	73	42	50	56	77	108	143	122	142

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.63	+0.76	N/A	N/A	N/A	N/A	N/A	-153	+256	+81

Nomenclature: See separate page